

Tianhao Li, Tianyu Liu, Wenbin Ge, Xiaodong Deng, Xiaohuan Zhou, Xingzhang Ren, Xinyu Zhang, Xipin Wei, Xuancheng Ren, Yang Fan, Yang Yao, Yichang Zhang, Yu Wan, Yunfei Chu, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zhihao Fan. 2024. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*.

Xin Zhang, Yanzhao Zhang, Dingkun Long, Wen Xie, Ziqi Dai, Jialong Tang, Huan Lin, Baosong Yang, Pengjun Xie, Fei Huang, Meishan Zhang, Wenjie Li, and Min Zhang. 2024. *mgte: Generalized long-context text representation and reranking models for multilingual text retrieval*. *Preprint*, arXiv:2407.19669.

Wenting Zhao, Xiang Ren, Jack Hessel, Claire Cardie, Yejin Choi, and Yuntian Deng. 2024. Wildchat: 1m chatgpt interaction logs in the wild. *arXiv preprint arXiv:2405.01470*.

Zhe Zhao, Hui Chen, Jinbin Zhang, Xin Zhao, Tao Liu, Wei Lu, Xi Chen, Haotang Deng, Qi Ju, and Xiaoyong Du. 2019. Uer: An open-source toolkit for pre-training models. *EMNLP-IJCNLP 2019*, page 241.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Tianle Li, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zhuohan Li, Zi Lin, Eric Xing, et al. 2023. Lmsys-chat-1m: A large-scale real-world llm conversation dataset. *arXiv preprint arXiv:2309.11998*.

Xuhui Zhou, Zhe Su, Tiwalayo Eisape, Hyunwoo Kim, and Maarten Sap. 2024. Is this the real life? is this just fantasy? the misleading success of simulating social interactions with llms. *arXiv preprint arXiv:2403.05020*.

Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, et al. 2023. Sotopia: Interactive evaluation for social intelligence in language agents. *arXiv preprint arXiv:2310.11667*.

Dmitry Zmitrovich, Alexander Abramov, Andrey Kalmykov, Maria Tikhonova, Ekaterina Taktasheva, Danil Astafurov, Mark Baushenko, Artem Snegirev, Tatiana Shavrina, Sergey Markov, Vladislav Mikhailov, and Alena Fenogenova. 2023. *A family of pretrained transformer language models for russian*. *Preprint*, arXiv:2309.10931.

A Contributions

Authorship order was determined randomly for all but the last author, as the paper was fully collaborative. We highlight the specific contributions made by individual authors.

Jonathan Ivey: Contributed to finding and developing some of the metrics used to measure text similarity, helped with project ideation and planning, and contributed to the creation of Figure 1.

Shivani Kumar: Contributed to identifying and implementing metrics used for text similarity, focusing on the lexical and syntactic aspects. Collaborated in creating the initial 50 prompts, assisted with annotations, and contributed to writing the paper (particularly §2 and §8).

Jiayu Liu: Contributed to evaluating the POTATO annotation tool. Assisted in manually annotating LLM simulation prompt development.

Hua Shen: Contributed to developing and analyzing the topic modeling of human conversation turns. Helped proposing and categorizing the metrics used for comparing humans and LLMs and partially contributed to writing the paper.

Sushrita Rakshit: Contributed to analyzing topic distribution outputs given by MALLETT. Assisted in manual annotation tasks and in developing initial prompts for LLM simulation. Also contributed to formatting dependent aggregate metrics and independent metadata into a large data file for regression model input; Assisted in hand-combing to remove multicollinear features and ran multivariate regression for aggregate scores (contribution to results in Figure 5).

Rohan Raju: Contributed to topic modeling and developing higher level topics, data cleaning for Russian LLM responses, and came up with several initial prompts. Helped developing experiment designs for human annotation tasks and annotating data. Worked on finding relevant papers for literature review (§2). Helped developing the regression tables and formatting them.

Haotian Zhang: Contributed to preprocessing and noise reduction in Wildchat data. Helped in building human annotation data. Helped in topic modeling.

Aparna Ananthasubramaniam: Contributed to project ideation and early literature review, procedures to standardize the metrics and make HUMAN/SIMULATOR comparisons, regression data preparation, and modeling (§7), setup of annotation task, development of early figures and appendix figures, and writing throughout the paper (particularly §4, §5, §6, §7, and supplement).

Junghwan Kim: Contributed to the design/implementation of lexical metrics, the visualization/interpretation/writing of regression results (§7), and the writing of multilingual results (§6). Helped with project ideation, annotation, code debugging, and writing in general.

Bowen Yi: Contributed to finding and developing all multilingual metrics and some lexical/style

metrics such as readability and toxicity. Helped with project ideation, literature review, data visualizations, and interpreting results (particularly for §7). Provided extensive human annotation and writing (particularly in §10 and §C).

Dustin Wright: Contributed to finding, developing, and orchestrating all of the metrics and measurements for the English data, wrote several of the initial 50 prompts, helped with dataset generation, helped with project ideation and scoping, and contributed to writing throughout the paper (particularly §3, §4, §5, and §7).

Abraham Israeli: Contributed to finding and developing some of the metrics used to measure text similarity. Led the literature review (§2) and helped with interpreting the regression results (§7). Contributed to writing throughout the paper, mainly §1, §2, §9, and §10.

Anders Giovanni Møller: Contributed to the design and implementation of the evaluation framework for English, Russian, and Chinese, including interpretation and analysis of the results, creation of visualizations and illustrations, and helped writing the paper (particularly §5 and §9).

Lechen Zhang: Contributed to preprocessing and sampling Wildchat data, collecting and constructing prompt dataset, generating and parsing all LLM simulated data, setting up the POTATO annotation tool, implementing and running multilingual evaluation metrics. Participated in writing the paper and creating visualizations (particularly §4 and correlation tables in the Appendix).

B Model Inference Details

Experiments are conducted on 8 NVIDIA RTX A6000 GPUs and 4 A100-SXM4-80GB GPUs using vLLM 0.5.4 (Kwon et al., 2023), Hugging Face Transformers 4.43.3 (Wolf et al., 2020) and PyTorch 2.4.0 (Paszke et al., 2019) on a CUDA 12.4 environment.

To ensure reproducibility, we set all random seeds in Python to be 1000, including PyTorch and NumPy. When doing model inference, we use temperature = 0.8, top_p = 0.95, and max_tokens = 1024.

C Annotation

We annotated 1,273 examples randomly sampled from the 2,000 examples in §5. This included a representative random sample of 863 examples used to calculate the F1 annotator baseline and an extra

upsample of 210 examples where HUMAN continued the conversation to increase the number of instances over which the linguistic features are compared.

Task Annotators are given the first turn of a dialogue between a HUMAN and the instructions from the top prompt in the DIRECT category. Annotators have to answer two questions: 1) whether the HUMAN will continue or end the conversation and 2) if the HUMAN continues the conversation, how they predict the HUMAN will respond. Annotations are conducted using POTATO (Pei et al., 2022). The annotation interface is pictured in Figure 8. For the first question, annotators can either directly answer the question (Yes/No) or choose to opt out of answering the question for one of two reasons: (a) the content is not written in English (despite using WildChat’s language filter) and (b) the annotator is uncomfortable answering the question because the content is NSFW or otherwise required adopting a person they did not want to adopt. Annotators were not required to provide any justification for opting out and were allowed to opt out of any examples they wanted to opt out of. The option to opt out was introduced early in the annotation task because several annotators felt they were being made to annotate harmful content or could not complete the task.

Sample We annotated 1,273 examples randomly sampled from the 2,000 examples in §5. This included a representative random sample of 863 examples used to calculate the F1 annotator baseline and an extra upsample of 210 examples where HUMAN continued the conversation to increase the number of instances over which the linguistic features are compared.

Output The annotation team consisted of 12 authors, including 11 university students and one faculty member. Of the 1,273 annotations, annotators selected "Yes" for 546 samples (43%) when they could directly answer the question and "No" for 542 samples (43%) when they could not. Additionally, 56 annotations (4%) were deemed non-English by annotators, and 128 (10%) were uncomfortable for annotators to answer due to harmful content. Neither of these two categories were considered in the annotator baseline for the experiment in §5. There were 293 cases where both the human annotator and HUMAN chose to continue the conversation, and this was the sample used to calculate the

linguistic metrics for §5.

D Regression Details for §7

Dependent Variable The dependent variable in the regression is the overall similarity between the HUMAN and SIMULATOR’s Turn 3 averaged across all measures $m \in M$ in Table 1. To calculate the similarity for each conversation, we first apply each measure m to pairs of HUMAN and SIMULATOR utterances for Turn 3, obtaining pairs (h_m, l_m) , and take one minus the distance between the pair. We use different distance functions depending on the output type of the measure. If m outputs a scalar, we use the absolute difference between h_m and l_m . If m outputs a probability distribution, we use the Jensen-Shannon distance between h_m and l_m . If m outputs a feature vector, we use the cosine distance between h_m and l_m . Using this, we obtain a vector of similarity scores s_m over all Turn 3 pairs. Additionally, to bring the metrics into approximately the same scale in order to be comparable and aggregate overall similarity across metrics, we log-scale the scalar metrics with an unbounded range which empirically demonstrates heavy-tailedness², followed by z-scoring each s_m . We then average the similarity scores s_m for all measures within each category to build four dependent variables corresponding to lexical, syntactic, semantic, and style similarity.

Independent Variables Our regression uses contextual features of the conversations obtained from the conversation metadata, the simulation metadata, and the conversation history (i.e., HUMAN message at Turn 1 and the LLM response at Turn 2). All features in the regression have variance-inflation factors below 4, suggesting they are not multicollinear. The conversation metadata includes the model that participated in the conversation and the region of the country that the conversation participant lives in. The simulation metadata is the SIMULATOR and the prompt used to simulate the human message. The conversation history is represented by a subset of scalar metrics that we used in §5. Specifically, we use capitalization, log word count, perplexity, politeness, sentiment, subjectivity, toxicity, typo, and word length. Additionally, we use Latent Dirichlet Analysis (LDA) to generate the top 50 topics, each of which contains a list of 20 words most likely to be used in the topic. We

further acquire the topic distributions for each of the human’s first turn in the input, and use these distributions as features in the regression, dropping the most common topic (“Information - research, social science”) to avoid collinear features. Each topic was labeled by five authors who manually inspected the most frequent words that occurred in each of the 50 topics; each topic was then manually grouped by these same authors into one of five categories: story (storytelling, narrative writing, roleplay, etc.), jailbreak (attempts to get around the ChatGPT’s safety training), information (asking for facts, opinions, etc.), programming (help with writing code), and other.

Regression coefficients are given in Table 9 and Table 10. p-values are corrected for multiple comparisons using a Bonferroni correction.

E Supplemental Results

Prompt Selection for §6 and §7 In order to select which three prompts were used in §6 and §7, two authors manually classified prompts into each category, and we selected one prompt per category. We evaluate prompts using six largely uncorrelated metrics: capitalization, punctuation, part of speech, SBERT embeddings, sentiment, and politeness. Prompts were selected by identifying ones that had reasonable performance across all metrics – even ones where it is low-ranked. Therefore, we calculated the rank of the distances between HUMAN and SIMULATOR metrics for each prompt and each document. We selected one prompt per category with the highest 75th percentile ranking, which tended to be prompts with high median rank and low variance across metrics (Figure 9). The three selected prompts are shown in Table 3.

Conversation End Prediction In addition to F1 scores, we evaluate how often each SIMULATOR predicts that a conversation will end as a function of whether the HUMAN actually ended the conversation in Figure 11. The same comparison is performed across models in Figure 12. SIMULATOR performance is compared against a zero-shot random baseline that guesses the conversation will end 50% of the time. Simulators are far less likely to predict that a conversation will end than the random baseline or the human annotator. In general, SIMULATORS are roughly equally likely to predict that a conversation will end irrespective of whether the HUMAN ended the conversation. This is true across models and languages. However, human

²This includes utterance length, average word length, perplexity, dependency tree depth, dependency tree breadth, and dependency tree distance.